

# LAB PRACTICAL-1

## DevOps Foundations & Continuous Integration

### 1. Check Git installation

```
Microsoft Windows [Version 10.0.19045.7548]
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C:\Users\sandy>git --version
git version 2.54.0.windows.1
```

### 2. Configure Git

```
C:\Users\sandy>git config --global user.name "N"manyasingh
C:\Users\sandy>git config --global user.email "manyasingh184@gmail.com"
C:\Users\sandy>git config --global --list
gui.recentrepo=C:/Users/sandy/OneDrive/Desktop/agentica ai/Super-250-Agentica-AI-Basics_OPEN_AI
user.manyasingh184@gmail.com
user.name=manyasingh
```

### 3. Create a GitHub repository and clone it

```
C:\Users\sandy>cd Documents
C:\Users\sandy\Documents>git clone https://github.com/manyasingh/capstone_devops.git
Cloning into 'capstone_devops'...
warning: You appear to have cloned an empty repository.
C:\Users\sandy\Documents>cd devops-demo
The system cannot find the path specified.
C:\Users\sandy\Documents>cd capstone_devops
```

### 4. Create a file

```
C:\Users\sandy\Documents\capstone_devops>echo "print('Hello DevOps')" > app.py
```

## 5. Check status

```
C:\Users\sandy\Documents\capstone_devops>git status
On branch main

No commits yet
```

## 6. Stage changes

```
C:\Users\sandy\Documents\capstone_devops>git add app.py
C:\Users\sandy\Documents\capstone_devops>git add .
```

## 7. Commit

```
C:\Users\sandy\Documents\capstone_devops>git commit -m "Add app.py with a hello message"
[main (root-commit) 472e550] Add app.py with a hello message
 1 file changed, 1 insertion(+)
 create mode 100644 app.py
```

## 8. Push to GitHub

```
C:\Users\sandy\Documents\capstone_devops>git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 84.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manyasingh/capstone_devops.git
 * [new branch]      main -> main
```

## 9. Pull latest changes

```
C:\Users\sandy\Documents\capstone_devops>git pull origin main
From https://github.com/manyasingh/capstone_devops
 * branch            main      -> FETCH_HEAD
Already up to date.
```

# Branching

## 10. Create a new branch and switch to it

```
C:\Users\sandy\Documents\capstone_devops>git checkout -b feature-login
Switched to a new branch 'feature-login'

C:\Users\sandy\Documents\capstone_devops>git switch -c feature-login
fatal: a branch named 'feature-login' already exists
```

## 11. Create/change a file

```
C:\Users\sandy\Documents\capstone_devops>echo "def login(): pass" > login.py
```

## 12. Add and commit

```
C:\Users\sandy\Documents\capstone_devops>git add login.py

C:\Users\sandy\Documents\capstone_devops>git commit -m "Add login function skeleton"
[feature-login 03327cc] Add login function skeleton
 1 file changed, 1 insertion(+)
 create mode 100644 login.py
```

## 13. Push the branch to GitHub

```
C:\Users\sandy\Documents\capstone_devops>git push origin feature-login
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 307 bytes | 13.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-login' on GitHub by visiting:
remote:   https://github.com/manyasingh/capstone\_devops/pull/new/feature-login
remote:
To https://github.com/manyasingh/capstone\_devops.git
 * [new branch]      feature-login -> feature-login
```

## 14. List branches

```
C:\Users\sandy\Documents\capstone_devops>git branch
* feature-login
  main
```

## 15. Switch back to main

```
C:\Users\sandy\Documents\capstone_devops>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

## 16. Merge the feature branch

```
C:\Users\sandy\Documents\capstone_devops>git merge feature-login
Updating 472e550..03327cc
Fast-forward
 login.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.py
```

## 17. Push merged changes

```
C:\Users\sandy\Documents\capstone_devops>git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manyasingh/capstone_devops.git
 472e550..03327cc  main -> main
```

## 18. Delete the branch

```
C:\Users\sandy\Documents\capstone_devops>git branch -d feature-login
Deleted branch feature-login (was 03327cc).
```

---



## Merge conflict

**:- After fixing the conflicting file**

```
C:\Users\sandy\Documents\capstone_devops>git add .

C:\Users\sandy\Documents\capstone_devops>git commit -m "Resolve merge conflict in app.py"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

**: - Git history**

```
C:\Users\sandy\Documents\capstone_devops>git log --oneline
03327cc (HEAD -> main, origin/main, origin/feature-login) Add login function skeleton
472e550 Add app.py with a hello message
```

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## Start Git in an existing project

**:-If you already have a project folder and want Git to track it**

```
C:\Users\sandy\Documents\capstone_devops>git init
Reinitialized existing Git repository in C:/Users/sandy/Documents/capstone_devops/.git/
```

**:- Then**

```
C:\Users\sandy\Documents\capstone_devops>git add .

C:\Users\sandy\Documents\capstone_devops>git commit -m "Initial commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

